Amjad S Althabteh

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Summary

C++ graphics and engine developers focused on real-time rendering and performance-critical systems. Built modular OpenGL engines, GPU-optimized pipelines, and custom allocators. Experienced with Unreal Engine 5 gameplay systems and low-level debugging.

Education

B.S in Computer Science GPA: 3.65/4.00 **Wayne State University** *Detroit, MI* ***2026***

Experience

Graphics Engineer, Co-Founder ***Immersalab Graphics*** *Detroit, MI* **Sep 2025 - Dec 2025**

- Architected a real-time C++20/OpenGL rendering engine with GPU-driven pipelines, emphasizing frame-time stability, cache efficiency, and performance-critical system design.
- Made an internal UI and JSON-based asset pipelines for flexible content creation.
- Implemented data-driven rendering optimizations for performance balancing.
- Added GPU profiling and frame-timing tools for debugging and optimization.

Unreal C++ Dev ***Indie Game (NOA)*** *Ann Arbor, MI* **Apr 2025 - Aug 2025**

- Developed gameplay systems in C++20 and visual scripting, integrating animation and narrative logic.
- Created and rigged 3D environments and characters.
- Produced a 90+ second real-time cinematic using MetaHuman, Control Rig, and Sequencer.
- Designed animation control systems with C++ and Animation Blueprints for smooth blending. *no_one_answers_slicee*

Software Development Intern ***Rare Munchies*** *Ann Arbor, MI* ***Sep 2023 - Apr 2024***

- Integrated UI components using **TypeScript**, improving page transitions and interaction flow.
- Identified and resolved **cross-browser compatibility issues**, reducing debugging time and unblocking delivery
- Optimized client-side rendering logic and state management patterns to reduce unexpected UI re-renders.

Research Experience

Undergraduate Researcher – Orbital Collision Modeling ***Wayne State University—Detroit, MI | Feb 2025 – Present***

- Modeled satellite orbital trajectories using numerical integration methods and simulated collision probabilities through N-body gravitation dynamics and spatial partitioning techniques.

Projects

3D Space Simulation Engine – C++/OpenGL *space-simulations*

- Engineered a custom 3D space simulation engine in C++17 using OpenGL 4.x, GLAD, and SFML.
- Developed modular engine architecture (rendering, physics, math, entities, and universe systems).
- Implemented custom math library (Vec3, Mat4) and camera system with projection/view matrices.

HFT OrderBook Engine C++ 20

- Architected a modular price-time FIFO matching engine with bid/ask depth tracking and partial fill execution logic.
- Designed low-latency infrastructure using cache-aligned data structures and modern C++20 features to minimize branch misprediction.

Glint 3D – C++ Automated 3D Rendering Engine *glint-rendering*

- Optimized a C++17/OpenGL rendering engine with optimized GPU workload and lighting.
- Cross-platform architecture with runtime scene analysis for stable performance.
- Implemented a custom C++20 memory pool allocator, reducing heap allocations and improving runtime stability under load.

C++ Runtime Stack Trace & Diagnostics Engine (Traceforge)

- Built a C++23 debugging system for automated stack-trace parsing.
- Automated regression test generation with intelligent fix suggestions.
- Added memory tracking utilities to detect leaks and invalid allocations.

Skill

- **Programming Languages**: **C++ (C++11–C++23), C, C#, Python, TypeScript, Java**
- **Graphics & Engines:** OpenGL • **Unreal Engine 4 - 5.7** • Unity • Godot • NX Design
- **Technical Specialities :** Gameplay Systems, VR/AR Development, Simulation, Tools & Plugins, Technical Art